

WEI Shengnan

The University of Macau
Faculty of Health Sciences

Phone: 0086-18811363326 / 00853-62892089
E-mail: yc27635@umac.mo / SWei.BME@outlook.com

Education

Aug. 2022 – present	The University of Macau	Macau SAR, China
	• Ph.D. student in Biomedical Sciences	
Aug. 2020 – Feb. 2022	The University of Sydney	Sydney, NSW, Australia
	• M. Eng. in Biomedical Engineering	
Aug. 2016 – Jul. 2020	Beijing University of Chemical Technology	Beijing, China
	• B. Eng. in Biomedical Engineering	

Research (Supervised by Prof. Guokai Chen at the University of Macau)

Aug. 2022 – present	Potential Impact of Early Fate Decision of hPSCs on the Tendency and Function of Offspring Cells.
	• Maintained human embryonic stem cell (hESC) lines H1 and H9. • Conducted research on the impact of various pathways on the transition from pluripotency to definitive endoderm. • Investigated the potential and tendencies of primitive streak subpopulations, focusing on varying phenotypes and profiles.
Dec. 2022 – present	Research on Key Pathways of <i>In Vitro</i> Induction and <i>In Vivo</i> Genesis of MSCs.
	• Conducted experiments to derive mesenchymal stem cells (MSCs) from pluripotent stem cells (PSCs) through different germ layers, utilizing both fetal bovine serum (FBS)-based and chemically defined methods. • Investigated the key components in traditional serum culture medium that contribute to the <i>in vitro</i> induction of MSCs. • Explored the potential improvements in MSC derivation by modifying the components of the traditional serum culture medium.
Apr. 2023 – present	Effect of Culture Conditions on Therapeutic Immunomodulation Ability of MSCs: A QC Study.
	• Cultivated various types of MSCs, including T-MSCs, BM-MSCs, UC-MSCs, AD-MSCs, and Colon-MSCs. • Conducted research to explore the effects of different components and supplements in various culture media on the immune regulatory function of MSCs. • Investigated the impact of different regulatory pathways on the immune regulatory ability of MSCs, aiming to enhance their therapeutic potential. • Implemented priming of MSCs with key cytokines such as interferon-gamma (IFN-γ), tumour necrosis factor-alpha (TNF-α), and interleukin-10 (IL-10) to modulate their immune regulatory properties.

Aug. 2022 – present	FEA Insights into the Mechanics of hPSC Colonies - Developed numerical models for human Pluripotent Stem Cell (hPSC) colonies. - Implemented qualitative Finite Element Analysis (FEA) methods for assessing the mechanics of hPSC colonies.
Research (Supervised by Dr Ali Entezari at the University of Sydney)	
Feb. 2021 – Dec. 2021	FEA study on porous total knee arthroplasty (TKA) prostheses - Developed numerical models for porous TKA prostheses. - Conducted FEA on TKA prostheses with varying porosity. - Optimized the design of porous components in TKA prostheses.
Research (Supervised by Prof. Xing Wang at Beijing University of Chemical Technology)	
Dec. 2020 – Feb. 2021	3D cell culture of mesenchymal stem cells (MSCs) with hydrogels - Cultivated mouse Mesenchymal Stem Cells (MSCs). - Investigated 3D culture conditions for mouse MSCs using Chitosan/di-terminal functionalized polyethylene glycol (DF-EPG) (CP) hydrogels.
Jun. 2019 – Jun. 2020	Adaptability study of CP hydrogels as 3D cell culture matrices - Synthesized DF-PEG from PEG. - Prepared CP hydrogels. - Adjusted the mechanical strength of the hydrogels to the desired levels. - Studied the impact of the mechanical strength of hydrogels on cell proliferation. - Optimized 3D culture conditions for L929 fibroblast cells and RSC96 Schwann cells.
Research (Supervised by Prof. Zheng-Jun Li at Beijing University of Chemical Technology)	
Jun. 2018 – Sept. 2018	Construction of genetically engineered bacteria - Cultivated genetically engineered bacterial strains. - Collected samples and characterized products using gas chromatography.
Publication	
Patent	Z. Li, W. Li, <u>S. Wei</u> , Q. Zhu, "Genetic engineering bacterium for simultaneously producing glycolic acid and lactic acid as well as construction method and application of genetic engineering bacterium", China Patent CN109337848A, Feb. 2019
Conference Presentations	
	<u>S. Wei</u> , Z. Zhang, G. Chen, "Qualitative analysis of hPSC colony mechanics with finite element method", Poster, 9th Macau Symposium on Biomedical Sciences (MSBS), Jun. 2023
	<u>S. Wei</u> , A. Entezari, "Design optimization of total knee arthroplasty prostheses: an FEA study", Poster, 7th Symposium on Biomedical Sciences for Students, Postdoctoral Fellows and Research Assistants, Apr. 2023
	<u>S. Wei</u> , Z. Zhang, G. Chen, "Qualitative analysis of hPSC colony mechanics with finite element analysis", Poster, 2nd International Guangdong-Hong Kong-Macau Greater Bay Area Conference in Regenerative Medicine (GBRM), Apr. 2023

S. Wei, A. Entezari, "Design optimization of total knee arthroplasty prostheses: an FEA study", Poster, 8th Macau Symposium on Biomedical Sciences (MSBS), Oct. 2022

Award

Sept. 2019	Best Student in the Advanced Practicum at China-Japan Friendship Hospital
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Skills

Experiments

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| • P1 and P2 Lab Work
●●●●● | • Cell Culture
●●●●● | • Flow Cytometry
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| • Molecular Cloning
●●●●● | • RT-qPCR
●●●●● | • 3D Modelling
●●●●● |
| • Finite Element Analysis
●●●●● | | |

Language

- Chinese: Native Speaker
- English: IELTS 7.0

Software

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|-----------------------|-----------------------|----------------------------|
| • SolidWorks
●●●●● | • Rhinoceros
●●●●● | • ANSYS Workbench
●●●●● |
| • Python
●●●●● | • Imaris
●●●●● | • MS Office
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Extracurricular activities

Aug. 2020 – Dec. 2021	Subcommittee member of the Sydney University Association of Biomedical Engineers, in charge of the WeChat Chinese student connection program and helping the organization of biweekly journal clubs.
Oct. 2016 – Oct. 2019	Member of the operation team for the official WeChat account of the College of Life Science and Technology, Beijing University of Chemical Technology, working on the design and editing of multimedia content and training of new members.
